

SOLVING TWO-STEP EQUATIONS

A step-by-step lesson with real-life examples and practice problems

Learning Objectives

- Understand what a two-step equation is
- Use the two-step strategy to solve equations
- Apply the method to real-life word problems
- Verify answers by substituting back

Section 1 — What Is an Equation?

Understanding the balanced-scale concept

An equation is like a **balanced scale** — both sides must always be equal. Whatever you do to one side, you **must do the same** to the other side to keep the balance.

$5 + 3 = 8$	$5 + 3 = 8$	8	✓
$2 \times 4 = 8$	$2 \times 4 = 8$	8	✓
$x + 2 = 9$	$x + 2$	9	Find x!

Section 2 — What Is a Two-Step Equation?

A **two-step equation** requires exactly **two operations** to isolate the variable. The general form looks like:

$$ax + b = c \quad \text{or} \quad x/a + b = c$$

Where **x** is the unknown, **a** is the coefficient, and **b** and **c** are numbers.

Section 3 — The Two-Step Strategy

Always work **backwards** using the **opposite (inverse)** operation:

Step	What to do	Opposite operation
1	Undo addition or subtraction	Subtract or Add
2	Undo multiplication or division	Divide or Multiply

Key rule: Whatever operation you apply to one side, you **must apply to both sides** to keep the equation balanced.

Section 4 — Worked Examples

Example 1 — Addition Type

$$3x + 5 = 20$$

Step 1: Subtract 5 from both sides: $3x + 5 - 5 = 20 - 5 \rightarrow 3x = 15$

Step 2: Divide both sides by 3: $3x \div 3 = 15 \div 3 \rightarrow x = 5$

Check: $3(5) + 5 = 15 + 5 = 20$ ✓

Answer: $x = 5$

Example 2 — Division Variable Type

$$x/2 + 7 = 12$$

Step 1: Subtract 7 from both sides: $x/2 + 7 - 7 = 12 - 7 \rightarrow x/2 = 5$

Step 2: Multiply both sides by 2: $(x/2) \times 2 = 5 \times 2 \rightarrow x = 10$

Check: $10/2 + 7 = 5 + 7 = 12$ ✓

Answer: $x = 10$

Example 3 — Subtraction Type

$$5x - 8 = 17$$

Step 1: Add 8 to both sides: $5x - 8 + 8 = 17 + 8 \rightarrow 5x = 25$

Step 2: Divide both sides by 5: $5x \div 5 = 25 \div 5 \rightarrow x = 5$

Check: $5(5) - 8 = 25 - 8 = 17$ ✓

Answer: $x = 5$

Section 5 — Real-Life Problems

■ The Toy Shop

Riya bought 2 toy cars and paid ₹30 extra for gift wrapping. Her total bill was ₹130. What is the cost of one toy car?

$$2x + 30 = 130$$

Solution:

Step 1: Subtract 30 from both sides $\rightarrow 2x = 100$

Step 2: Divide both sides by 2 $\rightarrow x = 50$

Answer: One toy car costs ₹50

■ The Fruit Seller

Amit bought 4 mangoes and spent ₹20 on a bag. His total bill was ₹100. What is the price of one mango?

$$4x + 20 = 100$$

Solution:

Step 1: Subtract 20 from both sides $\rightarrow 4x = 80$

Step 2: Divide both sides by 4 $\rightarrow x = 20$

Answer: One mango costs ₹20

■ The Birthday Party

Priya divided her chocolates equally among 3 friends and kept 4 for herself. She started with 16 groups' worth. How many chocolates altogether?

$$x/3 + 4 = 16$$

Solution:

Step 1: Subtract 4 from both sides $\rightarrow x/3 = 12$

Step 2: Multiply both sides by 3 $\rightarrow x = 36$

Answer: Priya started with 36 chocolates

■ The Book Fair

Rohan bought 6 notebooks and received a ■15 discount. He paid ■75 in total. What was the original price of one notebook?

$$6x - 15 = 75$$

Solution:

Step 1: Add 15 to both sides $\rightarrow 6x = 90$

Step 2: Divide both sides by 6 $\rightarrow x = 15$

Answer: One notebook originally cost ■15

Section 6 — Practice Time!

Try these on your own!

1	$2x + 6 = 14$	Addition type	
2	$4x - 3 = 13$	Subtraction type	
3	$x/5 + 2 = 7$	Division type	
4	$3x + 9 = 24$	Addition type	
5	$7x - 4 = 38$	Subtraction type	
6	$x/4 - 1 = 5$	Division/subtraction type	

Answers (check after attempting!)

Q	Answer	Q	Answer
1.	$x = 4$	4.	$x = 5$
2.	$x = 4$	5.	$x = 6$
3.	$x = 25$	6.	$x = 24$

Key Reminders

1. An equation is a **balanced scale** — both sides must stay equal.
2. Always undo **addition/subtraction first**, then **multiplication/division**.
3. Whatever you do to one side, **do the same to the other**.
4. **Always check** your answer by substituting back into the original equation.

You're an Algebra Star! Keep practising and have fun with maths!